

Task 1: Frequency, wavelength and velocity of propagation

$$\text{equation: } c = f \lambda = \frac{\lambda}{T}$$

$$c = 343 \text{ m/s}$$

calculate the missing quantities:

f	λ	T
440 Hz	0,5 m	
200 Hz		0,02 s
1000 Hz	1,7 m	
	2 cm	

Task 2: Multiple choice:

- ☐ A higher frequency means a longer wavelength.
- ☐ A higher frequency means a shorter wavelength.
- ☐ Frequency and wavelength are independent of each other.
- ☐ The speed of sound changes when the frequency changes.

Task 3:

A flute plays a note with a frequency of $f = 523 \text{ Hz}$

Calculate the wavelength of the sound in the air. Use $c = 343 \text{ m/s}$

Task 4:

A loudspeaker produces a sound wave with a wavelength of $\lambda = 2 \text{ m}$

Calculate its frequency.

Task 5:

A loudspeaker first plays a tone of 200 Hz and then a tone of 400 Hz.

For each quantity, discuss whether it increases, decreases or stays the same:

- frequency
- wavelength
- speed of sound

Task 6:

Two sound waves travel through air.

Wave A: 200 Hz

Wave B: 800 Hz

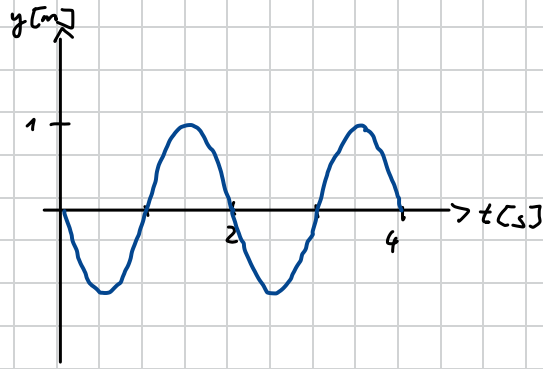
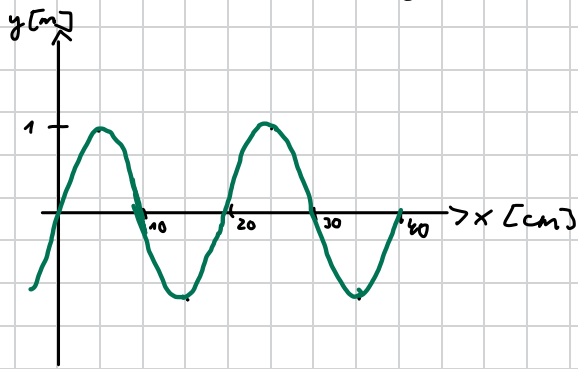
Discuss:

Which wave has the higher frequency?

Which wave has the longer wavelength?

Which wave travels faster?

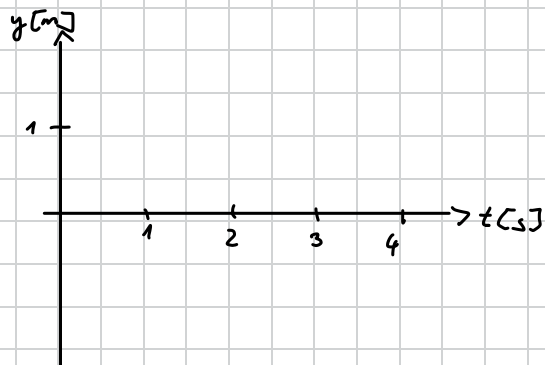
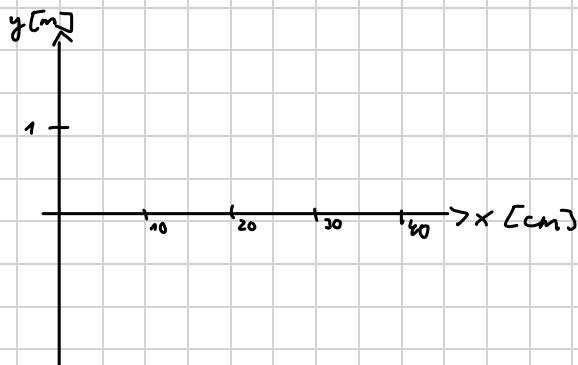
Task 7: Drawing waves
Consider the following wave



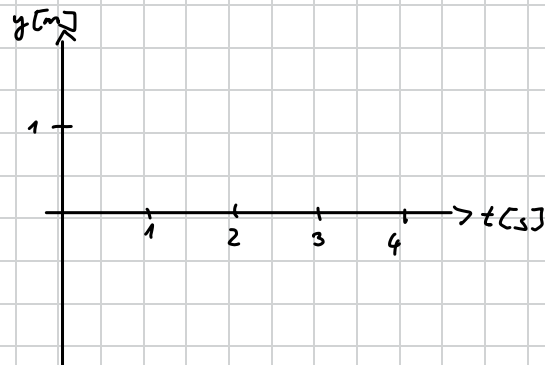
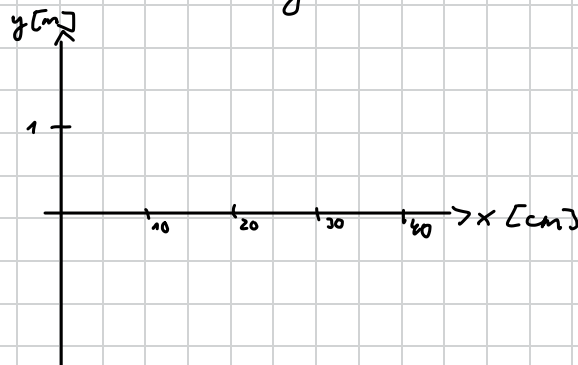
Discuss:

- Which graph shows the motion of one oscillator?
- Which graph shows the shape of the entire wave at one instant?
- What is shown on the horizontal / vertical axis of each graph?
- From which graph can you determine the period T ?
- From which graph can you determine the wavelength λ ?
- Determine c , λ , f and T

What changes if the amplitude is doubled? Draw the changes



What happens if c stays constant and f is increased up to $2f$.
Draw the changes



Harmonic Waves:

Read the following statements about a harmonic wave.
Decide whether the statements are True or False

- ☐ Every point oscillates harmonically
- ☐ The wave transports matter from one place to another.
- ☐ Every point oscillates with the same frequency.
- ☐ Neighbouring points have a phase difference.
- ☐ Every point has the same amplitude.

If a statement is false, discuss why.

Discuss the similarities and differences between travelling harmonic and standing waves

Standing Waves:

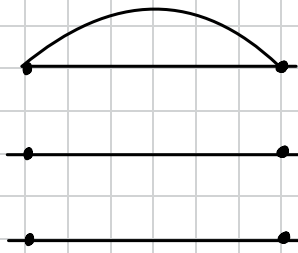
Draw the standing wave pattern for:

- the fundamental mode
- the first overtone
- the second overtone

Mark all nodes and antinodes.

Distinguish between two free ends, one free and one fixed end and two fixed ends.

two fixed ends



fundamental

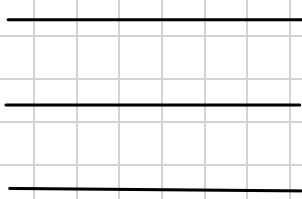
1st overtone

2nd overtone

different ends



two open ends



fundamental

1st overtone

2nd overtone

How are nodes and antinodes distributed. What is the distance between two nodes / antinodes

Consider a string with the length $L = 1,2\text{m}$ which is fixed at both ends. Calculate the wavelength of:

- the fundamental mode
- the first overtone
- the second overtone